**EC3040 Economics and Society**
**Assessment 2 – Coursework: Written assignment**

**Maria Klenina, 230015464**

## Part 1

### 1.1

### My prompt:

"Write a 300-word summary explaining the UK Labour Party's proposal to introduce a VAT on private school fees. Tell me how the policy works and include the expected costs and benefits it might bring through possible effects on school choices, funding of public schools, inequality, government revenue, and educational quality. You should use such economic concepts as externalities, redistribution, incentives, and behavioral responses".

### AI-generated response:

"The Labour Party proposes applying the standard 20% VAT rate to private school fees, which are currently exempt. The policy is intended to raise revenue and increase fairness in the education system. Adding VAT would make private school fees more expensive and could reduce demand for private education. Some families might switch to state schools, increasing enrolment pressures.

The policy's benefits include higher government revenue that can be reinvested in public schools, reducing inequality by improving the quality of state education. The VAT also aims to correct the implicit subsidy private schools receive by being VAT-exempt. Supporters argue that this would ensure private schools contribute fairly to the public system.

However, there are concerns. Some argue that reduced private school enrolment would increase class sizes and strain public school resources. Others worry the additional VAT costs may push middle-income families out of private education while wealthier families are unaffected, creating further polarisation. Critics also argue that private schools might offset the tax by cutting bursaries or reducing quality.

Overall, the VAT on private school fees aims to increase fairness, raise funds for state education, and reduce inequalities, but may also create risks for both systems depending on behavioural responses."

### 1.2
### Evaluation:

a) "The Labour Party proposes applying the standard 20% VAT rate to private school fees". – Correct, as IFS confirms that: "Labour has proposed levying VAT at the main rate of 20% on private school fees". (IFS 2023, page 4)

b) "The VAT aims to raise funds for state education". – It is right, as IFS supports this: "In the short run (one or two years), the impact on the level of private schooling is

likely to be small, with £1.5–1.6 billion representing the most likely net public finance impact in the short run". (IFS 2023, page 24)

c) "…the additional VAT costs may push middle-income families out of private education while wealthier families are unaffected." Underestimated by AI. IFS: "…a range of somewhere between a 3% and 7% fall in private school attendance (or about 20,000–40,000 pupils) as a result of a 15% effective VAT rate". (IFS 2023 page 26)

d) "The VAT also aims to correct the implicit subsidy private schools receive by being VAT-exempt". – Correct, but not enough, as IFS supports it: "Current tax exemptions could be seen as playing a similar role to such a voucher". "…there are also equity arguments for the state making a contribution to the cost of educating pupils across all sectors." (IFS 2023, page 6)

e) AI did not mention the equity argument (redistribution rationale). Lecture 9 on Education mentions that State education has equity and efficiency benefits: externalities, borrowing constraints and peer effects.

f) AI doesn't discuss the potential long-run school quality effects

g) AI doesn't mention anything about inelastic demand, as it is said in IFS: As we argue later, the demand for private schooling is likely to be inelastic – i.e. the number attending private schools is not that responsive to the price. The old Ramsey rule would suggest inelastic goods should be subject to higher-than-average tax rates to minimise deadweight costs and maximise revenues (Ramsey, 1927)". (IFS 2023, page 5)

**1.3**
**My summary:**

The Labour Party proposed to remove the current VAT exemption on private school fees and to apply the standard 20% rate. It is confirmed by the IFS, which states that "Labour has proposed levying VAT at the main rate of 20% on private school fees" (IFS, 2023, p. 4). The main fiscal motivation is to raise revenue for state education. In the short run, behavioural responses are expected to be limited, and IFS estimates "£1.5–1.6 billion, representing the most likely net public finance impact in the short run" (IFS, 2023, p. 24). These additional funds would go toward the state sector, where spending pressures are increased.

The policy might then affect household schooling decisions. The IFS states: a "3% to 7% fall in private school attendance (20,000-40,000 pupils)" can be expected in response to the effective VAT rate (IFS, 2023, p. 26), which gives more precise evidence. The resulting behavioural effect, though modest, may come with distributional consequences because rich households are less responsive to higher prices.

The IFS explains that "current tax exemptions could be seen as playing a similar role to such a voucher" and that there are "equity arguments for the state contributing to the cost of

educating pupils across all sectors" (IFS, 2023, p. 6). Therefore, eliminating the exemption improves redistributive goals and enhances the tax system's neutrality. Lecture 9 proves this by showing that state education generates equity and efficiency benefits through externalities, borrowing constraints, and peer effects.

There are also long-run effects on school quality and the role of inelastic demand that are worth talking about. The IFS also mentions that private schooling is "likely to be inelastic" and that, according to the Ramsey rule, such goods may efficiently be subject to higher taxation. Overall, the policy seems financially robust and driven by equitable considerations, though its wider educational effects rely on behavioural responses.

**Part 2**

2.1
**Public school**

The coursework defines the payoff from attending public school as:

$$U_i^{pub} = \frac{S}{x} + 5\,\frac{T}{x}$$

Public-school payoff

$U_i = 1000 - 100x$

where utility decreases with overcrowding

- $x$ : number of families attending public schools → overcrowding effect
- $\frac{S}{x}$ : baseline value of public education, decreasing in $x$
- $T$ : total VAT revenue raised from private schools
- $5\,\frac{T}{x}$ redistribution of VAT revenue to public schools.

$$U_i^{pub}(x) = \frac{S}{x} + 5 \times \frac{T}{x} = \frac{S + 5y\tau t}{x}$$

So, public-school utility increases when VAT revenue (T) is high and when there are less students attending public schools (x is lower).

**Private school**

Number of families that send their child to private school (number of private-school pupils):
y = 3 – x
Total fees: F = y*t

Benefits from private school:

$$u_i^{benefit} = P_i + 10\sqrt{\frac{F}{y}}$$

**Costs of private school:**

$(1+r)t$

Therefore, the payoff from attending privat school is:

$$① \quad u_i^{priv}(y) = P_i + \frac{10\sqrt{F}}{y} = P_i + \frac{10\sqrt{yt}}{y}$$

with $0 \leq P_L < P_M < P_H$, reflecting stronger preferences for private education among richer families

**2.2**
**Scenario 1: Only L in Public school, M and H in Private**

So:
x = 1
y = 2
F =2t
T = 2tτ

Total VAT revenue:

$T = \tau t y = 2\tau t$

Public school payoff

$U_{pub}(x) = \frac{S}{x} + 5\frac{T}{x} + u^{pub}(1) = S + 10\tau t$

Private school:

$\pi_i^{priv}(y) = P_i + \frac{10\sqrt{yt}}{y} - (1+\tau)t \quad , i \in \{L, M, H\}$

$y = 2 \Rightarrow \pi_i^{priv}(2) = P_i + \frac{10\sqrt{2t}}{2} - (1+\tau)t$ .

$\Rightarrow \pi_i^{priv}(2) = P_i + 5\sqrt{2t} - (1+\tau)t$

a) If L deviates from publ to priv $\Rightarrow y = 3$

$\pi_L^{priv}(3) = P_L + \frac{10\sqrt{3t}}{3} - (1+\tau)t$

So L stays in public if

$S + 10\tau t \geq P_L + \frac{10\sqrt{3t}}{3} - (1+\tau)t$

b) M doesn't deviate from priv to publ school

If M deviates then:

$x = 2, y = 1, T = \tau t$

$U_{pub}(2) = \frac{S}{2} + \frac{5\tau t}{2}$

So M stays public if

$P_M + 5\sqrt{2t} - (1+\tau)t \geq \frac{S}{2} + \frac{5\tau t}{2}$

c) H doesn't deviate from priv to publ school

Same deviation as for M

So, scenario 1 holds for all inequalities a,b,c

**2.3**

**Scenario 2: L and M in public school, H in private**

x = 2
y = 1

F = t
T = tτ

**Step 1: We compute utilities:**

Public school:

$$U_i^{pub} = \frac{S}{2} + 5\,\frac{t\tau}{2}$$

b) M doesn't deviate from public to private school

M stays public if

$$\frac{5}{2} + \frac{5\tau t}{2} \geq P_M + 5\sqrt{2t} - (1+\tau)t$$

c) H doesn't deviate from private to public school

if H deviates to public $\Rightarrow x=3, y=0$, $T = \tau t y = 0$

So H stays in private school if

$$P_H + 10\sqrt{t} - (1+\tau)t \geq \frac{5}{3}$$

So there is an equilibrium for scenario 2 if $(0 \leq P_L < P_M < P_H)$

Numerical example

$S=0$, $t=10$, $\tau=1$, $P_L=5$, $P_M=15$, $P_H=40$

$$U^{Pub}(2) = \frac{0}{2} + \frac{5\times1\times10}{2} = 25$$

$$\pi_L^{priv}(2) = 5 + \frac{10\sqrt{2\times10}}{2} - 20 \approx 7.36$$

$$\pi_M^{priv}(2) = 15 + \frac{10\sqrt{2\times10}}{2} - 20 \approx 17.36$$

$$\pi_H^{priv}(2) = 40 + \frac{10\sqrt{2\times10}}{2} - 20 \approx 54.62$$

$U^{Pub} = 0$ $(y=3)$

$\Rightarrow 57.62 \geq 0$

**2.4 How parameters affect scenarios**

- When ↑t or ↑τ: private schools become less attractive (More likely in scenario 2)
- ↑ S or ↑ redistribution (higher T/x): public schools attractiveness improves (More likely in scenario 2)
- ↑ Pi: there is a stronger preference for private schools (More likely in scenario 1)

**2.5 Effect of VAT on social welfare**

Social welfare:

$$W = U_L + U_M + U_H$$

1. Negative effect: higher fees of private-schools reduce utilities of private-school families
2. Positive effect: VAT makes T higher, increasing public-school utility

Welfare is increased by VAT if the benefit of redistribution exceeds:

- Overcrowding in public schools
- Forced transfer of families with a strong preference for private education

$U^{pub} = 0 \quad (y=3)$

$\Rightarrow 57.62 \geq 0$

Public school utility:

$U^{pub}(x) = \frac{5}{x} + \frac{5T}{x}, \quad T = \tau t y$

Private school utility:

$\pi_i^{priv}(y) = P_i + \frac{10\sqrt{yt}}{y} - (1+\tau)t$

If the family is in private school, the VAT payment changes with $\tau$, so

$$\frac{d\pi_i^{priv}}{d\tau} = -t$$

if family i is in public school

VAT affect utility through funding $T = \tau t y$, so

$$\frac{du^{pub}(x)}{d\tau} = 5\frac{dT/d\tau}{x} = 5\frac{ty}{x}$$

So, in any case each family from private school loses t from a marginal increase in $\tau$; each public family gains $\frac{5ty}{x}$

Also welfare effects of VAT increase ($x, y$ fixed)

When there are x public and y private $\Rightarrow x+y=3$

$$\Rightarrow \frac{dW}{dT} = x\left(\frac{5ty}{x}\right) + y(-t) = 5ty - ty = 4ty$$

Welfare can decrease when raising $\tau$ changes the equilibrium choices, then:

- The VAT base
- Changes because of changes in y
- Private-quality term

As $t>0$, $y \geq 0$

$$\frac{dW}{dT} = 4ty > 0 \quad \text{when } y>0$$

Scenario 1 $x=1$, $y=2$ (private M, H)

$$W_1 = 5 + (P_M + P_H) + 10\sqrt{2t} + 2t(4\tau - 1)$$

Scenario 2 $x=2$, $y=1$ (private H)

$$W_2 = 5 + P_H + 10\sqrt{t} + t(4\tau - 1)$$

$$W_1 \leq W_2$$

$$P_M \leq 10(\sqrt{2t} - \sqrt{t}) + t(1 - 4T)$$

In case of this inequality's failure, welfare reduces

### 2.6 Is VAT on private schools a good idea?

The desirability of the VAT, therefore, depends on the parameters of the model and how families respond to it. On the one hand, VAT could enhance equity and welfare effects because of inelastic demand for private education enrolment, as well as effective redistribution of VAT revenues generated, accruing benefits for public schooling. In the setup, VAT-generated revenues boost welfare from public schooling, as captured through 5T/x, benefiting those pupils in government schools because of better resources due to increased utility from public schooling. Because of much larger enrolment in public schooling, such redistribution would be beneficial for many pupils and help in making them more equally literate.

On the other hand, welfare gains from VAT will be diminished – or even negative – if VAT causes significant movement of pupils from private to public schools. When many families switch, the number of pupils in public schools rises, which reduces public-school utility through congestion because baseline quality and funding are spread over more pupils. In this case, congestion effects can dominate the gains from additional funding. Welfare effects are therefore ambiguous and depend on the elasticity of demand for private schooling, the strength of redistribution, and the capacity of the public education system. Consequently, VAT is not universally desirable, but its effectiveness depends critically on these underlying parameters.

## Part 3

### 3.1

In the theoretical model from Part 2 we get several clear and testable empirical predictions about the results of VAT introduction on private school fees.

**Prediction 1: VAT and private school attendance**

Raising the VAT rate on private school fees (t (1+ τ)) leads to a lower number and share of pupils in private schools. Higher prices of private schooling due to a higher VAT rate have a major effect on people who value private schooling least (low and middle-income families).

**Prediction 2: Public school funding and private school attendance**

An increase in public-school spending per student will lead to an increase in the advantage of public education and a subsequent reduction in private-school enrolment. The above can be represented in the model by an increase in S or an increase in VAT revenue per public-school student given by T/x.

**Prediction 3: Income and school choice**

Regions in which household income or wealth is high (Pi is high) have more pupils in private schools. Wealthy people value private education more and are not very sensitive to prices.

**Prediction 4: Diverse reactions to VAT**

The negative impact of VAT on private school enrollment is less in the case of high-income regions. This is because the demand for private education has less elasticity in the case of high-income classes, and the lower classes have a greater tendency to shift to public schools.

### 3.2 Measuring variables

**Attending private schools**
Expressed as the percentage of pupils who attend private (independent) schools.
Possible data sources:

- UK: School census data from Department for Education (DfE), or Independent Schools Council (ISC), by local authority or geographic area.
- The variables to use: Number of pupils in independent schools, Total number of pupils. SO, we can get Private school share (= independent school pupils / total pupils)

Source: https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25

Source: https://www.isc.co.uk/media/k5sfmobx/isc_census_2025-main-infographic.pdf

- International: The OECD "Education at a Glance" presents the percentage of enrollments in privately managed institutions.
- Variables to use: Percentage of students in private institutions (by country and education level)

Source: https://www.oecd.org/en/publications/education-at-a-glance-2025_1c0d9c79-en.html

**VAT and private school prices**

- VAT rate (τ): The statutory VAT rate that applies to private education (or Standard VAT Rate pre- and post-reform).
- Variables to use: Standard VAT rate (%) by country and year

Source: https://www.oecd.org/en/topics/taxation.html

- Tuition fees (t): Average private school fees using ISC data (UK) or education statistics.
- Variables to use: Average private school fees

Source: https://www.gov.uk/vat-rates

Source: https://www.isc.co.uk/research/annual-census/

**Public school funding/quality**
Stands for S and T/x

- UK: expenditure per pupil by local authority from DfE school funding statistics.
- Variables to use: Funding per pupil by local authority

Source: https://explore-education-statistics.service.gov.uk/find-statistics/school-funding-statistics/2024-25

- International: Public expenditure on educational institutions in OECD nations, by student.
- Variables to use: Public spending per student as a share of GDP

Source: https://data-explorer.oecd.org/vis?fs[0]=Topic%2C0%7CEducation%20and%20skills%23EDU%23&pg=0&fc=Topic&bp=true&snb=112&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_EAG_UOE_FIN%40DF_UOE_INDIC_FIN_PERSTUD&df[ag]=OECD.EDU.IMEP&df[vs]=3.1&dq=..ISCED11_1%2BISCED11_2%2BISCED11_3%2BISCED11_34%2BISCED11_35%2BISCED11_4%2BISCED11_5%2BISCED11_5T8%2BISCED11_6T8%2BISCED11_1T8._T.INST_EDU.DIR_EXP.V%2B_Z.USD_PPP_ST%2BPT_B1GQ_POP.&pd=2022%2C2022&to[TIME_PERIOD]=true

**Income**
Used as Proxy for (Pi)

- UK: Average Household Disposable Income by Region from Office for National Statistics.
- Variables to use: Mean disposable household income

Source: https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/incomeandwealth/datasets/nowcastinghouseholdincomeintheuk/current

- International: GDP per capita or household income data from the OECD.
- Variables to use: Gini (disposable income)

Source: https://data-explorer.oecd.org/vis?df[ds]=DisseminateFinalDMZ&df[id]=DSD_WISE_IDD%40DF_IDD&df[ag]=OECD.WISE.INE&dq=.A.INC_DISP_GINI..._T.METH2012.D_CUR.&pd=2010%2C&to[TIME_PERIOD]=false&vw=tb

### 3.3 Regression

Regression specification

We need to consider the following regression to test Prediction 2 (Public school funding and private school attendance):

$$\text{Private Share}_{r,t} = \alpha + \beta\ \text{Public Spend Per Pupil}_{r,t} + \gamma \log(\text{Income}_{r,t}) + \delta\ \text{Inequality}_{r,t} + u_{r,t}$$

where:

- $\text{Private Share}_{r,t}$ – the share of pupils in private ~~sho~~ schools in region r, in year t
- $\text{Public Spend per Pupil}_{r,t}$ measures state-school funding per pupil
- $\log(\text{Income}_{r,t})$ controls ability to pay for private schools
- $\text{Inequality}_{r,t}$ (for example Gini coefficient) shows distributional factors
- $u_{r,t}$ → error term

Predicted sign: $\beta < 0$

Higher spendings on public schools leads to public schools being more attractive, so it reduces private school attendance, as predicted by the model

### 3.4 Endogeneity concerns and solutions

Endogeneity is a key challenge in estimating the above regression

**Reverse causality:**

Going to a private school may have implications for government expenditures. For instance, if most children are attending private schools, then the government will pay less to fund public schools, resulting in biased estimations by OLS.

**Omitted Variables:**

Unobservable variables, such as the preference of the culture for private education and the tradition of schooling, can influence both public and private education.

**Simple errors:**

With public expenditure per student potentially measured with error, resulting in attenuation bias.

Possible solutions:

- Panel data with fixed effects, adjusting for time-invariant variables (regional characteristics).
- Difference-in-differences, increasing funding for public schools or imposing VAT on private school fees.
- Instrumental variables, through external changes in funding formulas or political variables that have an influence on the education spending but not on demand for private schools.

**Part 4**

Video 1 by Orchardpractice focuses on the potential short-term adverse effects of the introduction of VAT. The video suggests that raising the cost of private schools by 20% will have a particularly damaging impact upon middle-class families who, despite their economic circumstances, value private schooling despite the cost. Using the model developed in Part 2, this translates into a rise in the cost of private schooling of $t(1+\tau)$, reducing the utility of private schooling Pi for those with intermediate preferences Pi. The video's concern that schools will be full applies specifically to the congestion term of the payoff to attend a public school $(S/x) + 5(T/x)$: a significant number of pupils switching to public schools would cause x to rise, possibly degrading the quality of public schooling because of increased class sizes and a shortage of teachers. In this scenario, the introduction of VAT could worsen welfare outcomes if the resulting congestion effects exceed the benefits of redistribution.

Video 2 by LBCOfficial argues that VAT on private schools is a redistributive policy, but not just a "jealousy tax", as one of the debaters said. The presenter of the show states that there is only a small part of pupils leaving private schools and declining birth rates mean that there is enough capacity for everyone in public schools.
This is related to the empirical discussion in Part 3 and evidence from the IFS on limited switching. In the video, it is also mentioned that VAT revenue might improve the quality of teaching, facilities, and support for special needs in state schools. In the model, this is shown through an increase in T, which raises the utility offered by the public school via $5T/x$, benefiting many more children.
In conclusion, the videos demonstrate the core trade-off identified in coursework: while VAT could be applying short-term forces on state schools, with inelastic demand for private education and well-targeted redistributions of funds, it could lead to an overall improvement in equity provisions and even boost overall welfare.

**References:**


1. *Schools, pupils and their characteristics, academic year 2024/25* (no date) *GOV.UK*. Available at: https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25 (Accessed: 31 December 2025).
2. (No date) *Annual Census - ISC*. Available at: https://www.isc.co.uk/research/annual-census/ (Accessed: 21 December 2025).
3. (No date a) *Education at a glance 2025 | OECD*. Available at: https://www.oecd.org/en/publications/education-at-a-glance-2025_1c0d9c79-en.html (Accessed: 21 December 2025).
4. (No date a) *Taxation | OECD*. Available at: https://www.oecd.org/en/topics/taxation.html (Accessed: 21 December 2025).

5. (No date a) *Taxation | OECD*. Available at: https://www.oecd.org/en/topics/taxation.html (Accessed: 22 December 2025).
6. (No date a) *Annual Census - ISC*. Available at: https://www.isc.co.uk/research/annual-census/ (Accessed: 22 December 2025).
7. *School funding statistics, financial year 2024-25* (no date) *GOV.UK*. Available at: https://explore-education-statistics.service.gov.uk/find-statistics/school-funding-statistics/2024-25 (Accessed: 22 December 2025).
8. (No date a) *Data*. Available at: https://data-explorer.oecd.org/vis?fs%5B0%5D=Topic%2C0%7CEducation+and+skills%23EDU%23&pg=0&fc=Topic&bp=true&snb=112&df%5Bds%5D=dsDisseminateFinalDMZ&df%5Bid%5D=DSD_EAG_UOE_FIN%40DF_UOE_INDIC_FIN_PERSTUD&df%5Bag%5D=OECD.EDU.IMEP&df%5Bvs%5D=3.1&dq=..ISCED11_1%2BISCED11_2%2BISCED11_3%2BISCED11_34%2BISCED11_35%2BISCED11_4%2BISCED11_5%2BISCED11_5T8%2BISCED11_6T8%2BISCED11_1T8._T.INST_EDU.DIR_EXP.V%2B_Z.USD_PPP_ST%2BPT_B1GQ_POP.&pd=2022%2C2022&to%5BTIME_PERIOD%5D=true (Accessed: 23 December 2025).
9. Dataset(s): Nowcasting household income in the UK (no date a) Nowcasting household income in the UK - Office for National Statistics. Available at: https://www.ons.gov.uk/peoplepopulationandcommunity/personalandhouseholdfinances/incomeandwealth/datasets/nowcastinghouseholdincomeintheuk/current (Accessed: 23 December 2025).
10. (No date a) *Data*. Available at: https://data-explorer.oecd.org/vis?df%5Bds%5D=DisseminateFinalDMZ&df%5Bid%5D=DSD_WISE_IDD%40DF_IDD&df%5Bag%5D=OECD.WISE.INE&dq=.A.INC_DISP_GINI..._T.METH2012.D_CUR.&pd=2010%2C&to%5BTIME_PERIOD%5D=false&vw=tb (Accessed: 23 December 2025).